

UNIT-2: Probability Distributions — Identification Guide

◆ Overview

To solve problems in this unit, the **first and most critical step** is to correctly identify the type of distribution. Most questions can be solved easily once the correct model is chosen.

🎯 1. Binomial Distribution

✓ When to Use

- Number of trials (n) is fixed
- Each trial has **two outcomes** (success/failure)
- Probability (p) remains constant

🔑 Key Indicators

- “Out of n trials...”
- “Sample of size n ...”
- “Coins, boys/girls, defective items”

🧠 Recognition Rule

👉 If the problem involves **counting successes in a fixed number of trials**, use **Binomial**

🎯 2. Poisson Distribution

✓ When to Use

- Events occur randomly over time/space
- Average rate (λ) is given
- No fixed number of trials

🔑 Key Indicators

- “Average number per minute/day/week”
- “Number of accidents/calls/events”
- Rare events

🧠 Recognition Rule

👉 If the problem gives an **average rate of occurrence**, use **Poisson**

🎯 3. Normal Distribution

✓ When to Use

- Data is continuous
- Mean (μ) and standard deviation (σ) are given
- Distribution is symmetric (bell-shaped)

🔑 Key Indicators

- “Normally distributed”
- “Heights, weights, marks, wages”
- “Find proportion between...”

🧠 Recognition Rule

👉 If the problem involves **continuous data with mean & SD**, use **Normal**

🎯 4. Exponential Distribution

✓ When to Use

- Concerned with **time between events**
- Events occur continuously and independently

🔑 Key Indicators

- “Time until failure”
- “Waiting time”
- “Lifetime of a system/component”

🧠 Recognition Rule

👉 If the problem involves **waiting time**, use **Exponential**

⚖️ Quick Comparison Table

Feature	Binomial	Poisson	Normal	Exponential
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Nature	Discrete	Discrete	Continuous	Continuous
Key Input	n, p	λ	μ, σ	λ
Trials	Fixed	Not fixed	Not applicable	Not applicable
Use Case	Success count	Event rate	Measurement data	Waiting time

⚠ Important Distinction

♦ Binomial vs Poisson

- If n is small $\rightarrow$ Binomial
- If n is large & p is small $\rightarrow$ Poisson approximation

✖ Application to Your Question Set

Question	Distribution	Reason
Q1–Q4	Binomial	Fixed trials
Q5–Q10	Poisson	Average rate
Q11–Q14	Normal	Mean & SD
Q15	Fitting	Given frequency data

🧠 Final Identification Rule

👉 Read the **first line of the question**:

- “Out of / sample of” $\rightarrow$ **Binomial**
- “Average per...” $\rightarrow$ **Poisson**
- “Normally distributed...” $\rightarrow$ **Normal**
- “Time until...” $\rightarrow$ **Exponential**

✅ Conclusion

Accurate identification of the distribution type ensures:

- Correct formula usage
- Faster problem solving

- Higher accuracy in exams